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Chromosome sorting and PCR-based physical mapping in pea (*Pisum sativum* L.)

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Abstract

Three pea lines with reconstructed karyotypes were used for analysis and subsequent purification of individual chromosome types using flow cytometry and sorting. The lines JI 145, JI 146, and JI 148 possess defined chromosomal translocations allowing discrimination of three to four chromosome types from each line based on the different sizes of translocation chromosomes. Whereas only two chromosomes could be sorted from standard (wild-type) karyotype, a combined use of these lines allowed sorting of six out of the seven types of pea chromosomes. Chromosomes were identified and purity of flow-sorted fractions was assessed using fluorescence *in-situ* hybridization with a *PisTR-B* probe that was previously shown to give labelling patterns characteristic for each chromosome type. The fractions of flow-sorted chromosomes were of very high purity (> 95%) and proved to be suitable for detection of gene and marker sequences using PCR with specific primers. Three fractions containing chromosomes 2^7, 7^2 and a pool of all remaining chromosomes (1, 3, 4, 5, 6) flow-sorted from the line JI 148 were then used for PCR-based physical localization of genetic markers selected from linkage groups IV and VII. These experiments enabled assignment of the linkage groups IV and VII to chromosomes 4 and 7, respectively.

Introduction

Pea (*Pisum sativum* L., 2n = 14) is an economically important crop as well as the oldest model species in plant genetics (Mendel 1866). Several genetic maps of the pea genome have been constructed, using a variety of markers based on morphological, physiological and pigmentation characters, and isozyme or DNA polymorphisms (Weeden & Marx 1987, Ellis *et al*. 1992, Weeden *et al*. 1996, Gilpin *et al*. 1997, Hall *et al*. 1997b, Ellis *et al*. 1998, Laucou *et al*. 1998). The correspondence between linkage groups and chromosomes was investigated based on the behaviour of translocation stocks (Folkeson 1990a, Folkeson 1990b). A different approach, consisting of localizing genetically mapped rDNA, legumin, and vicilin genes on chromosomes *in situ* has recently been used by Fuchs *et al*. (1998) and has led to the assignment of most of the linkage groups from the map developed by Hall *et al*. (1997b) to individual chromosome types. Two linkage groups (IV and VII) remained unassigned

and were supposed to be located on either of the chromosomes 4 and 7. Although several morphological markers present in the linkage groups IV and VII in a composite map (Weeden *et al.* 1998) were previously assigned to chromosomes 7 and 4, respectively (Folkeson 1990b), these results remained to be confirmed (Weeden *et al.* 1996).

A convenient approach for physical mapping of genetic markers is the utilization of separated chromosomes. Although several methods of chromosome separation are available, flow sorting is probably the most efficient one. Localization of sequences on flow-sorted chromosomes can be performed either by PCR with specific primers or by membrane hybridization. Compared with membrane hybridization, PCR is more sensitive and requires a smaller number of chromosomes (Doležel *et al.* 1994). To date, this approach has been used only in *Vicia faba* to locate seed storage protein genes and in *Hordeum vulgare* to confirm the purity of flow-sorted chromosomes (Macas *et al.* 1993, Lysák *et al.* 1999). Since the pea chromosomes are similar in length, only two of its seven chromosome types can be separated from each other in the standard (wild-type) karyotype (Gualberti *et al.* 1996). This obstacle can be overcome by using lines with reconstructed karyotypes possessing defined translocations, as demonstrated in *Vicia faba* (Lucretti *et al.* 1993, Doležel & Lucretti 1995) and *Hordeum vulgare* (Lysák *et al.* 1999). Such lines were also described in pea (Lamm 1987, Simpson *et al.* 1990) and one of them (JI 146) has already been shown to be suitable for chromosome sorting (Neumann *et al.* 1998).

Here we demonstrate the use of three lines (JI 145 = L 83, JI 146 = L 84, JI 148 = L 108) with reconstructed karyotypes for chromosome separation by flow sorting. The peaks on relative fluorescence intensity histograms (flow karyotypes) were assigned to appropriate chromosomes using FISH with the *PisTR-B* probe, which allowed discrimination of all chromosomes of the standard karyotype (Neumann *et al.* 2001). Since the translocation of the line JI 148 was poorly defined, we also attempted to elucidate it on the basis of peak positions on flow karyotypes and *in-situ* hybridization results. Chromosomes sorted from the line JI 148 were used for assignment of linkage groups IV and VII to chromosomes using several PCR-based markers genetically mapped by Gilpin *et al.* (1997).

Material and methods

Plant material

Seeds of pea (*Pisum sativum* L., 2n = 14) lines with reconstructed karyotypes JI 145, JI 146 and JI 148 were kindly provided by Dr. Mike Ambrose (John Innes Centre, Norwich, UK). Seeds of pea cultivar 'Ctirad' with a standard karyotype were obtained from the plant breeding station SEMO (Smržice, Czech Republic).

Cell-cycle synchronization and chromosome isolation

Seed germination and all incubations were performed in the dark at 25±1°C and all solutions were aerated. To synchronize the cell cycle, pea seedlings with primary roots approximately 3 cm long were transferred onto a plastic tray filled with Hoagland's solution containing 1.25 mmol/L hydroxyurea (HU) for 18 h. Then the roots were washed in distilled water and incubated in HU-free Hoagland's solution for 3 h. To accumulate cells in metaphase, seedlings were transferred into 10 μmol/L amiprophos-methyl (APM) in Hoagland's solution for 2 h. Immediately after metaphase accumulation, the roots were cut 1 cm from the tip, rinsed in distilled water and fixed in 3% formaldehyde in Tris buffer (10 mmol/L Tris, 10 mmol/L Na_2EDTA, 100 mmol/L NaCl, pH 7.5) for 30 min at 5°C. The roots were washed three times for 5 min in Tris buffer and 25 root meristem tips (1 mm) were cut and transferred to a 5 ml polystyrene tube containing 1 ml LB01 lysis buffer (Doležel *et al.* 1989). Chromosomes were released mechanically using a homogenizer (Polytron PT 1200) at 13 000 rpm for 18 s. The suspension was passed through a 50-μm pore size nylon mesh to remove large cellular fragments.

Flow cytometry

Flow cytometric analysis and sorting was performed using the FACSVantage flow cytometer and sorter (Becton Dickinson, San José,

USA) equipped with an argon-ion laser tuned to multiline UV. A solution of 50 mmol/L NaCl was used as sheath fluid. The suspension of chromosomes was stained by DAPI (4′,6′-diamidino-2-phenylindole) at a final concentration of 2 μg/ml and analysed at rates of 200–400 particles per second. Relative fluorescence intensities, which corresponded to relative DNA content of chromosomes, were acquired on fluorescence 1 pulse area (FL1-A) histograms. Approximately 20 000–50 000 particles were analysed to generate flow karyotypes. To sort chromosomes, sorting windows were set on a dot-plot of FL1-A vs. FL1 pulse width (FL1-W).

Fluorescence in-situ *hybridization (FISH)*

FISH experiments were performed on chromosomes sorted on slides into LB01 buffer containing 5% sucrose. After air drying, the slides were stored at −20°C for up to two months. Just before hybridization, the slides were aged at 65°C for 45 min, washed 2 × 5 min in 2× SSC (1× SSC: 0.15 mol/L NaCl, 0.015 mol/L sodium citrate), fixed with 4% formaldehyde in 2×SSC (10 min), dehydrated in an ethanol series and air-dried. Hybridization probes for detection of *PisTR-B* (Neumann *et al.* 2001) and 5S rDNA sequences were labelled with Alexa Fluor 568-5- dUTP (Molecular Probes, Eugene, OR) or with biotin-16-dUTP (Boehringer Mannheim, Mannheim, Germany) using PCR with T3 and T7 primers and the corresponding plasmid clones as templates. Chromosome denaturation was carried out in 1×PCR buffer (Promega, Madison, WI) supplemented with 4 mmol/L $MgCl_2$ for 3 min at 94°C and followed by dehydration using an ice-cold ethanol series. Probe denaturation was performed in a hybridization mix lacking SSC for 15 min at 75°C. Then, the mix was chilled on ice and 20×SSC was added to obtain a complete hybridization solution [2×SSC, 50% formamide, 10% dextran sulphate, 125 ng/μl of sheared calf thymus DNA, 0.125% SDS (sodium dodecyl sulphate) and 0.025 ng/μl of biotin-labelled or 0.5 ng/μl of Alexa Fluor 568-labelled probe]. Hybridization with *PisTR-B* was done overnight at 28°C and post-hybridization washes were in 2×SSC at 32°C for 5 min and in 50% formamide in 2×SSC at 32°C for 10 min. Double-FISH experiments with *PisTR-B* and 5S rDNA probes were performed according to Neumann *et al.* (2001). Detection of biotinylated probes was as described by Leitch *et al.* (1994) using fluorescein–avidin DN and biotinylated anti-avidin D (Vector Laboratories, Houston, TX). The probes labelled with Alexa Fluor 568 were visualized directly. Chromosomes were counterstained with DAPI and examined using a Nikon Eclipse 600 microscope equipped with appropriate filter sets. The images were captured with a CCD camera and analysed using Lucia software (Laboratory Imaging, Prague, Czech Rep.).

Enumeration of translocation chromosomes followed the rule that the number refers to the chromosome providing the centromere, whereas the upper index indicates the second chromosome involved (Simpson *et al.* 1990).

PCR-based physical mapping

Three hundred chromosomes 2^7 and 7^2 or 1500 (5 × 300) chromosomes of the remaining types were flow-sorted from the line JI 148 into 30 μl of sterile water in PCR reaction tubes and stored at −70°C. Reaction mix (total volume 50 μl) consisted of 1×PCR buffer, 0.2 mmol/L dNTPs, 0.5 μmol/L primers, 2.5 mmol/L $MgCl_2$ and 2.5 U Platinum *Taq* DNA polymerase (Life Technologies, Gibco BRL, MD). Five primer pairs designed according to cDNA sequences genetically mapped by Gilpin *et al.* (1997) to linkage groups IV (I7 and Q500), VII (L58 and P628) and IB (Q407) were used (Table 1). Reaction conditions included initial denaturation (94°C, 5 min), 40 cycles of amplification (94°C, 1 min; 65°C, 1 min; 72°C, 2 min), and final extension (72°C, 10 min). PCR was carried out in a PTC-200 Peltier Thermal Cycler (MJ Research, Waltham, MA) and the reaction products were analysed on 1% agarose gels.

Results

Flow sorting and chromosome identification

The lines JI 145, JI 146 and JI 148 with reconstructed karyotypes were used for chromosome sorting. These lines possess reciprocal translocation T(3–5), T(3–6) and T(2–4 or 7),

Table 1. Characterization of the markers.

cDNA marker	Primer sequences 5′ → 3′	Expected length (bp)	Observed length (bp)	Linkage group*	Chromosome location
I7*	CTGATCATGTTCTTAAACTCGGTG CCAGATAGGGTAAACCAACAGAAG	1367	1367	VII	7
Q500	CTTTGTTTAGGTTGAAGGAAAACG CATAGGGTAACTCCACATCTCCTT	261	261	VII	7
L58*	GTTTCTGGTGGTGGAAGTGAAG AGGAATTTCAAGAGTCCAATTACG	406	406	IV	4
P628*	ATGTGGTCTCAACTGACTTTATTG AGCTATTGCAAACATGACTAAAC	640	640	IV	4
Q407	ACTTTGGAATGTCACTGAAATGG TAAGAAATGCTTGTTTGATTCTGC	242	350	IB	2

*Taken from Gilpin *et al.* (1997). Primer pairs for the remaining two markers were designed according to sequences Q500 and Q407 deposited in GenBank (accession nos. AA430919 and AA430905). Assignment of the linkage groups to individual chromosome types of the standard karyotype was deduced from PCR-localizations of the markers on chromosomes sorted from the line JI 148 (Figure 3). The linkage groups IV and VII of Gilpin *et al.* (1997) correspond to the linkage groups IVA and VII in the standard genetic map of Weeden *et al.* (1996). Since this map contains markers mapped by different authors, our assignment can be applied also to other maps, in which different linkage group designations are used (e.g. linkage group IVA of Weeden *et al.* (1996) corresponds to linkage group VII of Hall *et al.* (1997b); data not shown).

respectively (Lamm 1987, Simpson *et al.* 1990). High-resolution flow karyotypes were obtained after the analysis of chromosomes isolated from the three lines and the standard (wild-type) karyotype (Figure 1). The high quality of flow karyotypes was reflected by the very low coefficient of variation of chromosome peaks that ranged from 1.5 to 2%. In all cases, the translocations resulted in a rearrangement of peaks on flow karyotypes that allowed discrimination of more chromosome types than in the case of the line with the standard karyotype (Figure 1).

Chromosomes were sorted onto slides according to two parameters: fluorescence pulse area and fluorescence pulse width. The bivariate (FL1-A vs. FL1-W) sorting permitted the collection of high-purity fractions, in which contaminating chromosomes, chromatids, doublets or fragments of chromosomes were rarely observed. The chromosomes were identified by FISH with the satellite repeat *PisTR-B* (Figure 2). As described previously (Neumann *et al.* 2001), this probe allows discrimination of all chromosome types based on their signal patterns and morphology. The assignment of chromosome numbers to individual chromosome types of the standard karyotype was based on chromosome morphology and the co-localization of *PisTR-B* with 5S or 45S rDNA. However, it was difficult to discriminate between chromosomes 4 and 7, which are morphologically similar and no probe was available for their mutual discrimination. By a convention, chromosomes 4 and 7 carry a smaller and a larger satellite, respectively (Simpson *et al.* 1990). According to this presumption, satellite chromosome with three *PisTR-B* signals would be chromosome 4 and that with one signal chromosome 7. Moreover, chromosome 7 is somewhat larger than chromosome 4 (Hall *et al.* 1997a) which was reflected by its position on the standard flow karyotype (Figure 1).

In-situ hybridization with the *PisTR-B* probe allowed the identification of all chromosome types and confirmed the translocations described before (Lamm 1987, Simpson *et al.* 1990), except for chromosome 1 in the line JI 148, which lacked one *PisTR-B* signal and could not be discriminated from chromosome 7^2 (Figure 2). Taking advantage of the fact that the 5S rDNA genes are known to be located on chromosome 1 and are missing on chromosome 2 of the standard karyotype (Ellis *et al.* 1988), we performed double FISH with *PisTR-B* and 5S rDNA probes. 5S rDNA signals were obtained only on one chromosome type,

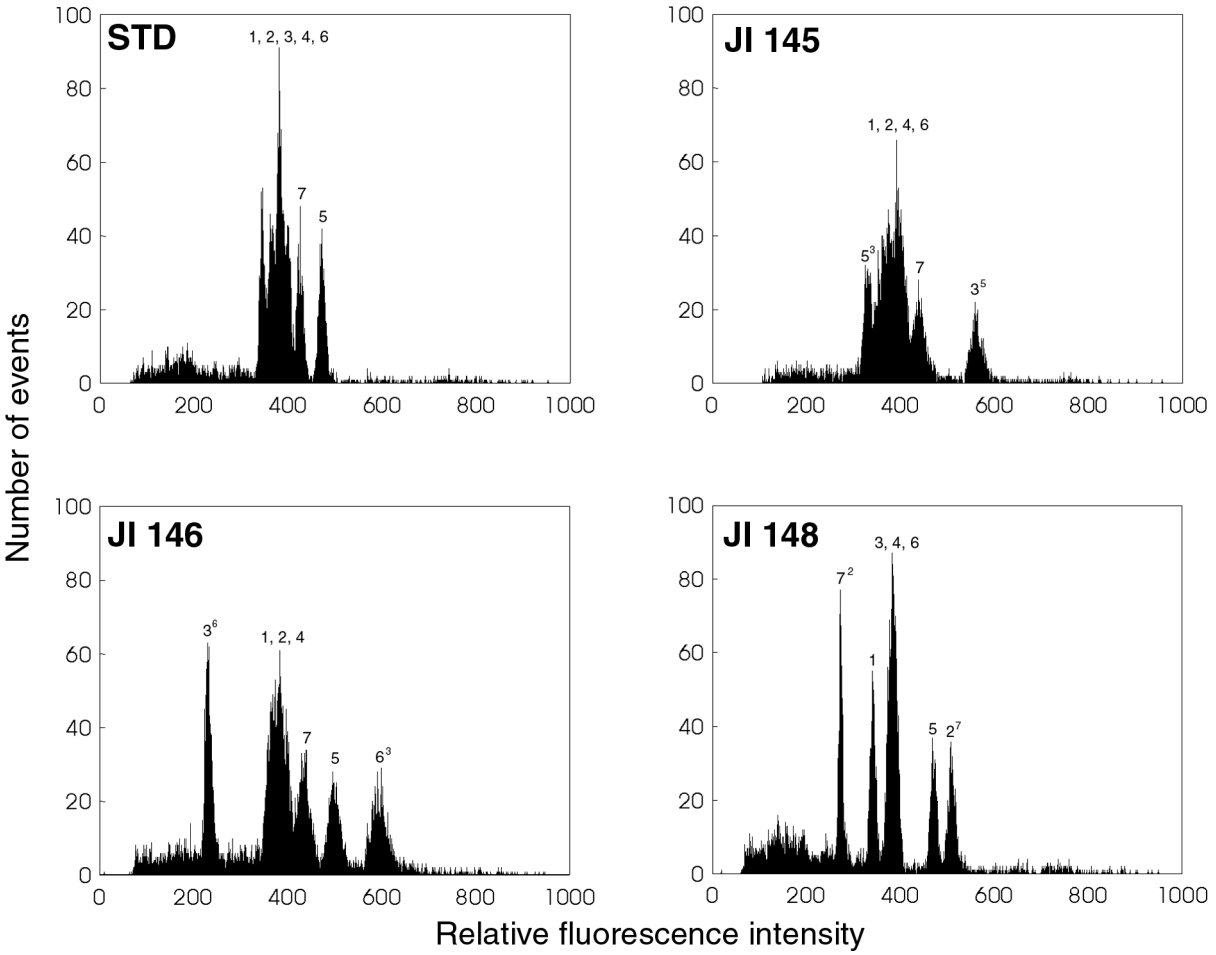
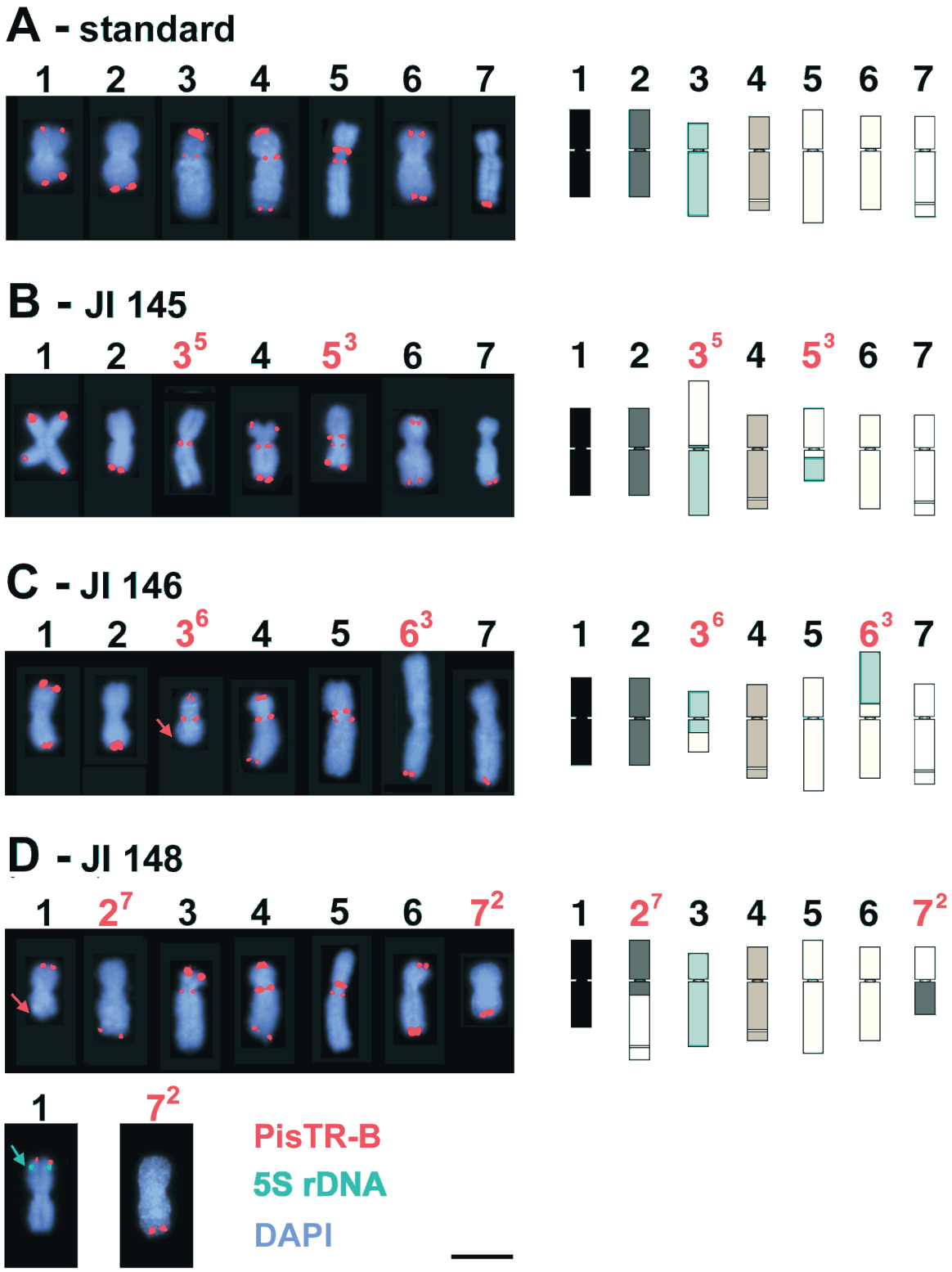


Figure 1. Flow karyotypes obtained after analysis of DAPI- stained chromosomes isolated from pea cultivar 'Ctirad' with a standard karyotype (STD) and from three lines with reconstructed karyotypes (JI 145, JI 146 and JI 148). Assignment of individual peaks to chromosomes was made using *in-situ* hybridization with the *PisTR-B* probe performed on sorted chromosomes (Figure 2).

on the same arm as *PisTR-B* (Figure 2D). Thus the chromosome with the 5S rDNA signal was identified as chromosome 1. The absence of one *PisTR-B* signal was also found on chromosome 3^6 in the line JI 146 (Figure 2C). In spite of that, this chromosome type could be discriminated from others in this line. We attempted to detect the missing *PisTR-B* loci on chromosomes 1 and 3^6 in the lines JI 148 and JI 146 using more sensitive detection of a biotinylated *PisTR-B* probe. However, this experiment also failed to generate the expected signals.

The chromosomal translocation in the line JI 148 has been poorly defined, as it was not known which of chromosomes 4 and 7 is involved in the translocation (Lamm 1987, Simpson *et al.* 1990). Furthermore, the involvement of chromosome 2 in the translocation was not absolutely clear because the translocation was reported to be genetically linked with marker 'p' from linkage group VI (Lamm 1983), which was assigned to chromosome 1 by Fuchs *et al.* (1998). Since two of the peaks on the flow karyotype of JI 148 correspond to translocation chromosomes originating from chromosomes 2 and 7 of the standard karyotype, we can conclude that the translocation in this line occurred between chromosomes 2 and 7. The shorter one of the two translocation chromosomes is clearly metacentric and is shorter than chromosome 2 of the standard karyotype


A - standard
1 2 3 4 5 6 7
1 2 3 4 5 6 7
B - JI 145
1 2 3[5] 4 5[3] 6 7
1 2 3[5] 4 5[3] 6 7
C - JI 146
1 2 3[6] 4 5 6[3] 7
1 2 3[6] 4 5 6[3] 7
D - JI 148
1 2[7] 3 4 5 6 7[2]
1 2[7] 3 4 5 6 7[2]
1
7[2]
PisTR-B
5S rDNA
DAPI

(Figures 1 & 2D). Furthermore, the short arm of chromosome 7 is shorter than that of chromosome 2 in the standard karyotype (Hall *et al.* 1997a). If the translocation is reciprocal (Lamm 1987, Simpson *et al.* 1990), this means that the shorter translocation chromosome contains the centromere of chromosome 7 and can be designated as 7^2 whereas the longer one is 2^7.

PCR-based physical mapping

Three fractions containing chromosomes 2^7, 7^2 and the remaining chromosomes pooled together were sorted from the line JI 148 and used for PCR localizations of genetic markers from the linkage groups IV and VII described by Gilpin *et al.* (1997). The markers were selected with the aim to assign these linkage groups to chromosomes 4 and 7. PCR with primers for I7 and Q500 markers selected from the linkage group VII resulted in products of the correct length in sorted fractions corresponding to chromosomes 7^2 and 2^7, respectively (Figure 3). This suggests that the linkage group VII is located on chromosome 7, and that the Q500 marker is present on a segment involved in the translocation with chromosome 2 (Table 1). Consequently, the linkage group IV was expected to correspond to chromosome 4. This assumption was confirmed by detection of P628 and L58 markers selected from this group only in the composite fraction containing all chromosomes except 7^2 and 2^7 (Figure 3). In addition to these four markers, we also localized the marker Q 407 from linkage group IB of Gilpin *et al.* (1997). This linkage group corresponds to linkage group II of Hall *et al.* (1997b), now generally called linkage group I (Laucou *et al.* 1998, Fuchs *et al.* 1998), that was assigned to chromosome 2 by Fuchs *et al.* (1998). The marker was localized on chromosome 2^7, confirming the identity of this chromosome (Figure 3).

Discussion

In this study, we demonstrated that the use of pea translocation lines JI 145, JI 146 and JI 148 allowed sorting of more chromosome types than the line with the standard karyotype. The combination of these three lines allowed discrimination and sorting of all pea chromosomes, with the exception of chromosome 4. However, it should also be possible to sort this chromosome using lines with suitable translocations that have already been described (Klein 1973, Lamm 1987).

The flow-sorted fractions corresponding to individual peaks from all tested lines were characterized using FISH with the *PisTR-B* probe (Neumann *et al.* 2001) in order to identify chromosome types present in these fractions. In the standard and JI 146 karyotypes, the previously published identification of chromosomes based on chromosome lengths and theoretical flow karyotypes (Gualberti *et al.* 1996, Neumann *et al.* 1998) has been confirmed. In all lines with reconstructed karyotypes the expected translocations have been detected, although chromosomes 3^6 in JI 146 and 1 in JI 148 lacked one *PisTR-B* signal. Since the positions of peaks on the flow karyotypes were as expected and the correct chromosome identification was proved by using another probe (Figure 2), this finding indicates variability in the number of *PisTR-B* signals among these lines rather than the wrong assignment of the translocations or the occurrence of unexpected chromosome rearrangements. This is also supported by similar variability in the number of hybridizing loci of satellite repeats in other plant species (Houben *et al.* 2000, Shibata *et al.* 2000).

The translocation in the line JI 148 was elucidated as T(2–7). Involvement of the satellite chromosome (7) is in agreement with the data of Lamm (1983) and Simpson *et al.* (1990). On the other hand, the involvement of chromosome 2 is in contradiction to supposed linkage between the translocation point and marker 'p' (Lamm

Figure 2. Fluorescence *in-situ* hybridization using *PisTR-B* probe on the chromosomes flow-sorted from standard (**A**) and reconstructed karyotypes (**B–D**). All chromosomes were unambiguously identified by specific labelling patterns, except for the line JI 148 where double FISH with *PisTR-B* and 5SrDNA probes was carried out to distinguish between chromosomes 1 and 7^2. The sites of *PisTR-B* hybridization are labelled in red, 5SrDNA signals are blue-green, and the positions of missing *PisTR-B* signals are marked by red arrows. All karyotypes are schematically depicted on the right side of the figure. Bar = 10 μm.

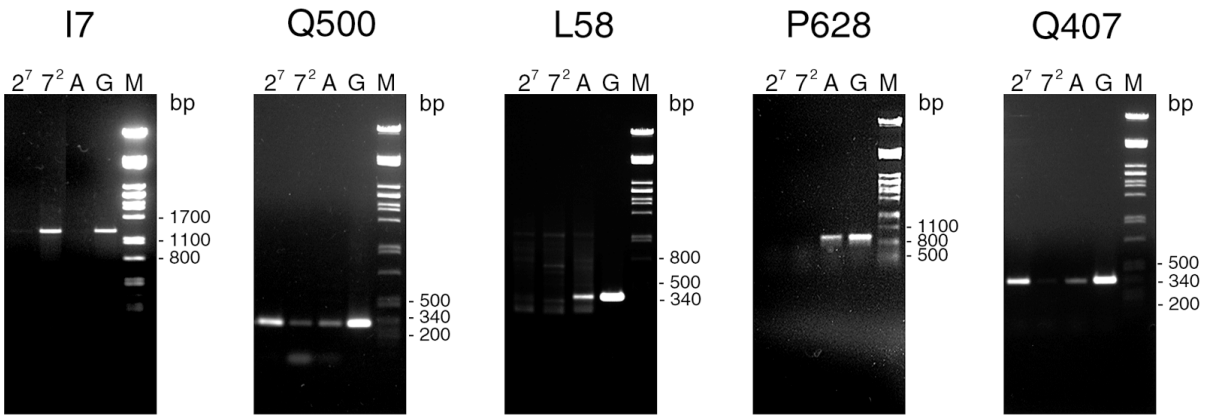


Figure 3. PCR-based physical mapping of genetic markers using chromosomes flow-sorted from the pea line JI 148. The reactions were performed separately on 300 chromosomes 2^7, 7^2 and 300 chromosomes of each of the five remaining types pooled together (lanes A). Genomic DNA (2.5 ng) was used as a positive control (lanes G). PCR products were resolved on 1% agarose gel electrophoresis. Lambda DNA digested with *Pst*I was used as a size marker (lanes M).

1983). This marker is associated with linkage group VI and hence chromosome 1 (Fuchs *et al*. 1998). However, it should be noted that the linkage distance reported by Lamm (1983) was rather large. Our evidence showing that chromosome 2 is involved in the translocation is based on clear discrimination of chromosomes and on the results of physical mapping using markers selected from linkage groups IB, IV and VII. Moreover, our data are consistent with Simpson *et al*. (1990) who also concluded that chromosome 2 is involved in the translocation.

This study marks the completion of the assignment of all genetic linkage groups to individual chromosomes in *P. sativum*. The assignment of linkage group IV to chromosome 4 and group VII to chromosome 7 described in this paper is in contradiction with the results of Folkeson (1990b) who assigned these linkage groups to the same chromosomes but in the opposite pairwise combination. However, as pointed out by Weeden *et al*. (1996), there appeared to be some confusion in the identification of the two satellite chromosomes, because they are difficult to distinguish using only cytogenetic approaches. Our results are supported by PCR-localization of four markers selected from these linkage groups and exact identification of individual chromosome types by *in-situ* hybridization with *PisTR-B* probe and by high-resolution flow karyotyping.

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